

D 110209

(Pages : 3)

Name.....

Reg. No.....

**FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2024**

Mathematics

MTS 5B 06—BASIC ANALYSIS

(2020 Admission onwards)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

Answer any number of questions.

Each question carries 2 marks.

Maximum 25 marks.

1. Prove that there does not exist a rational number r such that $r^2 = 2$.
2. Determine the set A of $x \in \mathbb{R}$ such that $|2x + 3| \leq 7$.
3. If $a, b \in \mathbb{R}$, prove that $|a + b| \leq |a| + |b|$.
4. State the supremum property of $\mathbb{R}$.
5. If $S = \left\{ \frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N} \right\}$, find $\inf S$ and $\sup S$.
6. State and prove Archimedean property.
7. Let x and y be real numbers with $x < y$, prove there exists an irrational number z such that $x < z < y$.
8. State and prove squeeze theorem.
9. If a sequence (x_n) of real numbers converges to a real number x , prove that any subsequence (x_{n_k}) of (x_n) also converges to x .

Turn over

10. Prove that every Cauchy sequence of real numbers is bounded.
11. Let (x_n) and (y_n) be two sequence of real numbers and suppose that $x_n \leq y_n$ for all $n \in \mathbb{N}$. If $\lim x_n = +\infty$, prove that $\lim y_n = +\infty$.
12. Prove that the intersection of any finite collection of open sets in $\mathbb{R}$ is open.
13. Compute $(1 + \sqrt{3}i)^9$.
14. Find the real and imaginary parts of $f(z) = z^2 - (2 + i)z$ as a function of x and y .
15. Show that the complex function $f(z) = z + 3i$ is a one to one on the entire complex plane and find a formula for its inverse function.

Section B

Answer any number of questions.

Each question carries 5 marks.

Maximum 35 marks.

16. State and prove Cantor's theorem.
17. Let a and b be positive real numbers, prove that $\sqrt{ab} \leq \frac{a+b}{2}$ and the equality occurs if and only if $a = b$.
18. State and prove density theorem.
19. Prove that unit interval $[0, 1]$ is not countable.
20. State and prove monotone convergence theorem.
21. Let $F \subseteq \mathbb{R}$; prove that the following are equivalent :
 - (a) F is a closed subset of $\mathbb{R}$;
 - (b) If $X = (x_n)$ is any convergent sequence of element in F , then $\lim X$ belongs to F .
22. Find an upper bound for $\left| \frac{-1}{z^4 - 5z + 1} \right|$ if $|z| = 2$.

23. For any *two* complex numbers, prove that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$.

Section C

*Answer any **two** questions.
Each question carries 10 marks.*

24. a) If A_m is a countable set for each $m \in \mathbb{N}$, prove that $A = \bigcup_{m=1}^{\infty} A_m$ is countable.
b) State and prove Bernoulli's inequality.
25. a) State and prove monotone convergence theorem.
b) Let $s_1 = 1$ and $s_{n+1} = \frac{1}{2} \left(s_n + \frac{a}{s_n} \right)$ for $n \in \mathbb{N}$. Prove that (s_n) converges to $\sqrt{a}$.
26. a) Prove that every contractive sequence is a Cauchy sequence.
b) Let $f_1 = 1, f_2 = 1$ and $f_{n+1} = f_n + f_{n-1}$. Define $x_n = \frac{f_n}{f_{n+1}}$. Prove that $\lim x_n = \frac{-1 + \sqrt{5}}{2}$.
27. Find a complex linear function that maps the equilateral triangle with vertices $1 + i, 2 + i$ and $\frac{3}{2} + \left(1 + \frac{1}{2}\sqrt{3}\right)i$ onto the equilateral triangle with the vertices $i, \sqrt{3} + 2i$ and $3i$.

(2 × 10 = 20 marks)